FAILURE OF UPPER SEMICONTINUITY OF HILBERT–KUNZ MULTIPLICITY ON A REDUCED NON-EQUIDIMENSIONAL RING

ARGUS OPEN RESEARCH PROJECT

Abstract. Problem 1.5 of the AIM problem list on test ideals and multiplier ideals asks to investigate the upper semicontinuity of $\mathfrak{p} \mapsto e_{\mathrm{HK}}(R_{\mathfrak{p}})$ on $\operatorname{Spec} R$, without imposing a local-equidimensionality hypothesis. Over $k = \mathbb{F}_p$, we prove that the reduced finite-type ring $R = k[a,b,c,d]/(ab, acd)$ has prime ideals $\mathfrak{p} = (b,c,d) \subsetneq \mathfrak{m} = (a,b,c,d)$ satisfying $e_{\mathrm{HK}}(R_{\mathfrak{p}}) = 2 > 1 = e_{\mathrm{HK}}(R_{\mathfrak{m}})$, so the sublevel set $\{\mathfrak{q} \in \operatorname{Spec} R : e_{\mathrm{HK}}(R_{\mathfrak{q}}) < 3/2\}$ is not Zariski open.

1. Exact Monomial-Length Calculation and Proof

Let $k = \mathbb{F}_p$ ($p > 0$ prime), $A = k[a,b,c,d]$, $I = (ab, acd)$, and $R = A/I$.

Theorem 1.1. *In $R = k[a,b,c,d]/(ab,acd)$:*

(1) *$I = (a) \cap (b,c) \cap (b,d)$ is an irredundant intersection of prime ideals of dimensions $3, 2, 2$, so R is reduced, excellent, and F-finite, but not locally equidimensional at $\mathfrak{m} = (a,b,c,d)$.*

(2) *At the prime $\mathfrak{p} = (b,c,d) \subsetneq \mathfrak{m}$, $a \notin \mathfrak{p}$ is a unit, so $R_{\mathfrak{p}} \cong k(a)[c,d]_{(c,d)}/(cd)$ has dimension 1, standard monomial basis $\{1, c, \dots, c^{q-1}, d, \dots, d^{q-1}\}$ modulo $\mathfrak{p}^{[q]}$ of length $2q-1$ for every $q = p^e$, and Hilbert–Kunz multiplicity*

$$e_{\mathrm{HK}}(R_{\mathfrak{p}}) \;=\; \lim_{q \to \infty} \frac{2q-1}{q^1} \;=\; 2.$$

(3) *At the maximal ideal $\mathfrak{m} = (a,b,c,d)$, $\dim R_{\mathfrak{m}} = 3$, and counting standard monomials $a^i b^j c^r d^s$ ($0 \le i, j, r, s < q$) avoiding ab and acd gives q^3 monomials with $i = 0$ and $(q-1)(2q-1)$ monomials with $1 \le i < q$ and $j = 0$, for a total length of $q^3 + 2q^2 - 3q + 1$ and Hilbert–Kunz multiplicity*

$$e_{\mathrm{HK}}(R_{\mathfrak{m}}) \;=\; \lim_{q \to \infty} \frac{q^3 + 2q^2 - 3q + 1}{q^3} \;=\; 1.$$

Since $\mathfrak{p} \subsetneq \mathfrak{m}$ and $e_{\mathrm{HK}}(R_{\mathfrak{p}}) = 2 > 1 = e_{\mathrm{HK}}(R_{\mathfrak{m}})$, the function $\mathfrak{q} \mapsto e_{\mathrm{HK}}(R_{\mathfrak{q}})$ is not upper semicontinuous on $\operatorname{Spec} R$.

Date: August 27, 2026.

2020 *Mathematics Subject Classification.* Primary 13A35; Secondary 13D40.